



**can you help me fill in the theoretical framework section of my thesis? Provide academic research papers/sources from 2021 - 2025 ONLY. My thesis topic is Early Detection of the Risk of Diabetes Using Calibrations on Explainable Artificial Intelligence (XAI) with Philippine Data.**

Diabetes risk prediction using calibrated XAI is well-aligned with current research trends in diabetes epidemiology, machine learning for tabular clinical data, and reliability-focused evaluation in healthcare from 2021–2025.<sup>[1] [2]</sup>

## **A. Diabetes and Prediabetes**

Diabetes mellitus is a chronic metabolic disease characterized by hyperglycemia due to defects in insulin secretion (type 1) or insulin action and secretion (type 2), leading to microvascular and macrovascular complications such as nephropathy, retinopathy, and cardiovascular disease. Prediabetes represents an intermediate state of dysglycemia (often defined by impaired fasting glucose or impaired glucose tolerance) where blood glucose is elevated but below diagnostic thresholds for diabetes, yet already associated with increased cardiometabolic risk. In the Philippines, an estimated 4.3 million adults had diagnosed diabetes and 2.8 million remained undiagnosed in 2021, while recent national surveillance suggests that roughly 20% of adults aged 20–59 are prediabetic, underscoring a large pool of high-risk individuals who could benefit from early risk prediction.<sup>[3] [4] [5] [1]</sup>

Early prediction is essential because Filipino adults with diabetes show high rates of complications such as diabetic nephropathy contributing to around one-third of renal disease cases and a substantial share of preventable blindness due to diabetic retinopathy, meaning delayed detection translates into dialysis, disability, and premature death. Studies based on national nutrition and aging cohorts demonstrate that obesity, metabolic syndrome, sedentary behavior, and poor diet significantly increase the odds of young-onset and late-onset diabetes among Filipinos, indicating that modifiable risk factors emerge years before clinical diagnosis. Predictive models that flag high-risk individuals during this prediabetic window can support targeted lifestyle and pharmacologic interventions, which Philippine practice guidelines increasingly emphasize as cost-effective strategies for curbing diabetes burden.<sup>[4] [6] [7] [3] [1]</sup>

Philippine-specific patterns matter because genetic, sociocultural, and environmental determinants of diabetes in Filipinos differ from those in Western populations commonly used to train risk scores. Genomic work has identified variants in candidate genes that significantly

alter type 2 diabetes susceptibility among Filipinos, while community studies show distinct profiles of diet, physical activity, and health beliefs in rural versus urban settings that shape risk trajectories. Recent national and regional surveys also document rising prevalence in older adults and unique barriers to care during events such as the COVID-19 pandemic, reinforcing the need for prediction models calibrated on Philippine datasets rather than imported tools whose risk thresholds and features may not transfer well.<sup>[8] [9] [6] [10] [4]</sup>

## B. Machine Learning and Binary Classification

Diabetes risk prediction is naturally modeled as a binary classification problem because individuals are labeled as “at risk/likely to develop diabetes” versus “not at risk,” often based on glycemic cut-offs or incident diabetes within a follow-up period.<sup>[2] [11]</sup> Many studies train classifiers on structured clinical variables such as age, BMI, blood pressure, lipids, and family history, with models outputting a probability  $P(y = 1|x)$  that can be thresholded to assign each patient to the positive (high-risk) or negative class.<sup>[11] [12]</sup> Some frameworks extend this to multi-class outcomes (e.g., normal, prediabetes, diabetes), but core formulations and most evaluation metrics remain grounded in binary discrimination between high-risk and low-risk states.<sup>[11] [13]</sup>

Machine learning models learn patterns in such data by optimizing a loss function over large sets of labeled examples, adjusting parameters so that the predicted risk probabilities align with observed outcomes. In recent work, deep architectures for tabular data, gradient boosting machines, and logistic regression have all been shown to capture non-linear interactions among risk factors better than traditional scores, yielding higher accuracy and area under the ROC curve when predicting type 2 diabetes onset. Importantly, these models can ingest high-dimensional EHR features, genomic markers, and behavioral variables, making them suitable for rich Philippine datasets that combine clinical, lifestyle, and socio-economic indicators.<sup>[12] [2] [14] [8]</sup>

For health-risk prediction, classification metrics that emphasize positive-class performance are critical because missing high-risk patients has greater clinical cost than flagging some low-risk individuals. Several diabetes prediction studies therefore report recall (sensitivity), F1-score, or F2-score, showing that models optimized for recall can achieve sensitivities above 85–90% while maintaining acceptable specificity, which is particularly important in screening contexts. Precision–recall curves and area under the precision–recall curve (AUPRC) are also recommended for imbalanced datasets, where diabetes cases form a minority, since AUPRC better reflects performance on the positive class than ROC-AUC.<sup>[11] [15] [16] [2] [12]</sup>

## C. Calibration Theory

Calibration in probabilistic classification refers to the agreement between a model’s predicted probabilities and the true empirical frequencies of events, such that among patients assigned a 0.7 diabetes risk, roughly 70% actually develop diabetes. Well-calibrated models support downstream decision-making because clinicians can map probability thresholds to actions (e.g., further testing, lifestyle program enrollment) with predictable trade-offs, whereas overconfident or underconfident models mislead users about the strength of evidence. In healthcare, poor calibration can result in inappropriate reassurance or unnecessary

interventions, making calibration a core requirement for models deployed in clinical practice and regulatory evaluation.<sup>[17] [16] [18] [19]</sup>

Probability reliability is particularly essential in medicine because treatment decisions, resource allocation, and patient communication often depend on risk thresholds rather than mere class labels. Recent work on medical imaging and clinical prediction shows that even highly accurate deep learning systems can be substantially miscalibrated, with confidence far exceeding true accuracy in certain subgroups or operating regions. Integrating calibration objectives or explicit uncertainty quantification into training has been shown to reduce expected calibration error and provide more reliable confidence estimates, thereby improving safety and clinician trust.<sup>[16] [18] [19] [17]</sup>

Classical post-hoc calibration methods include Platt scaling, which fits a logistic mapping from raw scores to calibrated probabilities, and temperature scaling, which rescales logits using a single temperature parameter optimized on validation data. Extensions such as spline-based or isotonic regression methods, and train-time approaches that incorporate calibration losses like average calibration error as auxiliary objectives, have been demonstrated to meaningfully reduce miscalibration without severely degrading discrimination. These methods are directly applicable to diabetes risk prediction models, allowing probabilities produced by TabNet or other classifiers to be tailored to Philippine prevalence and risk distributions.<sup>[19] [17]</sup>

Calibration also corrects overconfident models by penalizing situations where high predicted probabilities are assigned to incorrect predictions, nudging the model toward probability estimates that better match observed outcomes. In medical applications, auxiliary calibration losses or confidence-weighted penalties have reduced calibration error by sizable margins (e.g., 15–20% relative decreases) while preserving or slightly improving accuracy in tasks such as cardiac MRI classification and segmentation. For a thesis on early diabetes detection, this means calibrated XAI models can deliver risk scores that are not only discriminative but also numerically meaningful for Filipino patients.<sup>[20] [17] [19]</sup>

## Clinical importance of calibrated probabilities

Calibrated probabilities matter in clinical contexts where downstream decisions are threshold-based, such as starting preventive medications, ordering confirmatory tests, or prioritizing patients for limited lifestyle programs. Studies in laboratory medicine and AI regulation highlight that clinical validation now increasingly requires demonstration of not only discrimination but also calibration, especially when models guide high-stakes decisions like chronic disease screening. In the Philippine setting, where resources are constrained and prevalence is high, calibrated risk predictions can help set context-appropriate thresholds that balance sensitivity with cost and avoid overburdening the healthcare system with false positives.<sup>[7] [18] [1] [16]</sup>

## D. TabNet and Optimization Strategies

TabNet is a deep neural architecture specifically designed for tabular data that uses sequential attention to select relevant features at each decision step, combining the strengths of decision trees and neural networks. In diabetes prediction, TabNet has achieved robust performance compared to traditional models, with studies reporting competitive or superior accuracy and AUC while retaining the ability to provide feature-level importance through its attentive masks. Because electronic health records and survey-based Philippine datasets are predominantly tabular, TabNet is well-suited as a backbone model for the thesis. [\[2\]](#) [\[14\]](#) [\[21\]](#)

Research on explainable diabetes classification using a Bayesian-optimized TabNet architecture shows that hyperparameter optimization (e.g., via Bayesian optimization) can significantly improve predictive performance on structured clinical datasets. For instance, a TabNet model tuned with Bayesian optimization has been reported to reach high accuracy and F1-score on Pima and clinical diabetes datasets, outperforming baseline deep learning and tree-based methods. Optimization strategies such as Bayesian optimization or swarm-based metaheuristics reduce the number of expensive training runs needed to find high-performing hyperparameters, which is important when experimenting with calibration losses and XAI pipelines. [\[13\]](#) [\[21\]](#) [\[2\]](#)

## E. Explainable Artificial Intelligence (LIME and SHAP)

Interpretability is crucial in medicine because clinicians must understand why a model predicts high diabetes risk before integrating its output into clinical judgment and patient counseling. Black-box predictions without explanation can undermine trust, hinder error detection, and conflict with ethical and regulatory expectations for transparent decision support systems. XAI techniques such as LIME and SHAP help bridge this gap by exposing feature contributions and supporting human-AI collaboration in clinical risk assessment. [\[15\]](#) [\[18\]](#) [\[11\]](#) [\[16\]](#)

LIME provides local explanations by fitting an interpretable surrogate model (often linear) around each individual prediction, highlighting which features increased or decreased a specific patient's predicted diabetes risk. Studies applying LIME to diabetes datasets report that local explanations enable clinicians to verify that high-risk predictions align with known risk factors like high BMI, hypertension, and dyslipidemia, thereby improving acceptance of ML-based screening tools. In TabNet-based frameworks for diabetic risk prediction, LIME has been used to generate patient-level explanations that complement model-intrinsic feature masks and make outputs understandable to non-technical users. [\[21\]](#) [\[11\]](#) [\[13\]](#)

SHAP, based on Shapley values from cooperative game theory, offers both local and global explanations by attributing each feature's marginal contribution to the prediction across many coalitions of features. Recent comparative studies show that SHAP can provide consistent global importance rankings and rich visualizations that reveal non-linear effects and interaction patterns in diabetes risk models. When integrated with TabNet or other high-capacity models, SHAP plots have been used to show, for example, how increasing age or fasting glucose drives predicted risk, which helps clinicians validate and refine risk stratification strategies. [\[11\]](#) [\[15\]](#) [\[13\]](#)

Clinicians tend to trust interpretable models more when explanations match medical knowledge and when uncertainty in predictions is communicated transparently. Work combining LIME and SHAP in diabetes prediction reports improved clinician understanding and greater willingness to rely on model outputs for decision support, because both local and global views of feature importance were available. In a calibrated XAI setting, coupling trustworthy probabilities with clear feature attributions can make model recommendations actionable in Philippine clinics and community programs.<sup>[18] [7] [1] [16] [13] [11]</sup>

## F. Model Performance and Evaluation Metrics

The F2-score is a variant of the F-measure that weights recall more heavily than precision, which is appropriate when failing to detect high-risk patients is more serious than over-referral. In recent diabetes prediction models, emphasizing recall through metrics like F2 has enabled researchers to tune decision thresholds that achieve high sensitivity (often above 90%) while maintaining reasonable specificity, making these models more suitable for population screening rather than definitive diagnosis. For Philippine datasets with high undiagnosed prevalence, a recall-oriented metric aligns with public health goals of catching as many at-risk individuals as possible.<sup>[5] [1] [15] [12] [2] [11]</sup>

Precision–recall curves and their area (AUPRC) are particularly informative in imbalanced medical datasets, where positive cases (diabetes or high risk) are much less frequent than negatives. Studies in medical AI show that models with similar ROC-AUC can differ substantially in AUPRC, with calibrated and well-tuned models achieving higher precision at clinically relevant recall levels, which better reflects real-world screening performance. Using AUPRC alongside ROC-AUC thus provides a more realistic assessment of how the model will behave when deployed in Philippine populations where diabetes prevalence is moderate but still lower than the non-diabetic majority.<sup>[10] [15] [16] [12]</sup>

These metrics fit imbalanced medical data because they focus on true positives and false negatives (recall) and the proportion of predicted positives that are correct (precision), rather than being dominated by the majority class. In diabetes risk prediction, an evaluation framework that prioritizes F2-score, recall, and AUPRC—combined with calibration metrics such as expected or average calibration error—captures both the ability to detect high-risk Filipinos and the reliability of the associated probabilities. Such a framework provides a rigorous theoretical basis for the thesis, connecting epidemiologic urgency, ML methodology, calibration theory, and explainability into a coherent model for early detection using Philippine data.<sup>[7] [1] [17] [15] [16] [2]</sup>

# for the machine learning models i will be using (Logistic Regression, SVM, Random Forest, KNN, XGBoost) and for Calibrations i will be using (Platt scaling and temperature scaling)

Those model and calibration choices fit your topic very well. Below is wording you can integrate into your Theoretical Framework under “Machine Learning and Binary Classification” and “Calibration Theory,” focused on Logistic Regression, SVM, Random Forest, KNN, XGBoost, and Platt/temperature scaling.

## Machine learning models

For early diabetes risk prediction using structured clinical and lifestyle data, this thesis employs a diverse set of supervised classification algorithms—Logistic Regression, Support Vector Machines (SVM), Random Forests, k-Nearest Neighbors (KNN), and Extreme Gradient Boosting (XGBoost)—each capturing different patterns in tabular health indicators. Logistic Regression provides a probabilistic baseline with interpretable coefficients that quantify how risk factors such as age, BMI, hypertension, and fasting glucose linearly influence the log-odds of diabetes, and it has achieved accuracies above 90% in several recent diabetes datasets, demonstrating that even simple linear decision boundaries can be effective when predictors are well chosen. SVM extends this by finding a maximum-margin separating hyperplane, often in a nonlinear kernel-induced feature space, and has been reported to reach high accuracy and sensitivity in early-stage diabetes detection, particularly when class imbalance is addressed and informative kernels are selected.<sup>[40] [41] [42] [43]</sup>

Ensemble and instance-based models further enhance predictive capacity by leveraging non-linear relationships and local structure in the data. Random Forests aggregate many decision trees trained on bootstrap samples and random feature subsets, yielding robust models that handle complex feature interactions and have produced diabetes prediction accuracies in the mid-80% to mid-90% range across multiple studies. KNN, in contrast, assigns risk based on the labels of the most similar patients in feature space and has shown competitive performance for early diabetes prediction when coupled with appropriate feature scaling and distance metrics, making it a useful comparator that reflects local neighborhood structure in Philippine clinical data. XGBoost, a gradient-boosted tree algorithm, has consistently outperformed many single models for diabetes prediction, with recent work reporting near-perfect accuracy or top-ranked AUC among compared algorithms, which motivates its use as a strong non-linear baseline in this thesis.<sup>[41] [44] [42] [43] [40]</sup>

## Platt scaling and temperature scaling

Because the thesis focuses not just on classification but on trustworthy risk probabilities, the raw outputs of these models are post-processed using two parametric calibration techniques: Platt scaling and temperature scaling. Platt scaling fits a logistic regression model that maps each classifier’s scores or margins to calibrated probabilities, and recent methodological work shows that, when applied on a held-out validation set, it can notably reduce expected

calibration error while preserving discrimination, even in high-dimensional settings. Temperature scaling, originally popularized for deep neural networks, rescales logits or decision scores by a single temperature parameter optimized to minimize a calibration loss; medical AI studies report that this simple transformation can substantially improve probability reliability without degrading ROC-AUC, particularly for overconfident models.<sup>[45]</sup>  
[46] [47] [48] [49] [50]

Contemporary guidance for clinical prediction models emphasizes that evaluation must include both discrimination and calibration, recommending simple recalibration techniques—such as intercept/slope adjustment, Platt scaling, or temperature scaling—before models are considered for deployment. Recent cost-aware extensions of these methods further demonstrate that tuning Platt or temperature parameters with respect to asymmetric misclassification costs can yield probabilities that are better aligned with clinical decision thresholds, which is particularly relevant in diabetes screening where missing high-risk patients is more harmful than over-referral. By applying Platt and temperature scaling to Logistic Regression, SVM, Random Forest, KNN, and XGBoost trained on Philippine data, this thesis aims to produce calibrated risk estimates that are numerically meaningful for clinicians, enabling threshold selection based on local prevalence and resource constraints while maintaining strong predictive performance.<sup>[47]</sup> <sup>[51]</sup> <sup>[50]</sup> <sup>[52]</sup> <sup>[44]</sup>

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1. <https://pubmed.ncbi.nlm.nih.gov/38382166/>
2. <https://dl.acm.org/doi/10.1016/j.compbiomed.2022.106178>
3. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8214347/>
4. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9758548/>
5. <https://www.abs-cbn.com/news/health-science/2025/6/6/2-in-10-pinoy-s-are-prediabetic-study-1520>
6. <https://pmc.ncbi.nlm.nih.gov/articles/PMC11163309/>
7. <https://www.sciencedirect.com/science/article/abs/pii/S1871402124000122>
8. <https://pmc.ncbi.nlm.nih.gov/articles/PMC11575783/>
9. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10805280/>
10. <https://idf.org/our-network/regions-and-members/western-pacific/members/the-philippines/>
11. <https://www.diva-portal.org/smash/get/diva2:1886442/FULLTEXT02.pdf>
12. <https://www.nature.com/articles/s41598-025-20534-4>
13. <https://www.tandfonline.com/doi/full/10.1080/10255842.2025.2566962?src=>
14. <https://pmc.ncbi.nlm.nih.gov/articles/PMC12256431/>
15. <https://www.sciencedirect.com/science/article/pii/S2772442525000097>
16. <https://www.sciencedirect.com/science/article/pii/S0893395225000018>
17. <https://arxiv.org/html/2506.03942v1>
18. <https://www.tandfonline.com/doi/full/10.1080/10408363.2025.2488842?af=R>
19. <https://dl.acm.org/doi/10.1145/3690624.3709232>

20. <https://pmc.ncbi.nlm.nih.gov/articles/PMC12115728/>
21. <https://research.usq.edu.au/item/z022x/explainable-diabetes-classification-using-hybrid-bayesian-optimized-tabnet-architecture>
22. image.jpg
23. image.jpg
24. <https://academic.oup.com/jes/article/doi/10.1210/jendso/bvaf149.1045/8299152>
25. <https://ijmaberjournal.org/index.php/ijmaber/article/view/2667>
26. <https://academic.oup.com/jes/article/doi/10.1210/jendso/bvaf149.1097/8299473>
27. <https://academic.oup.com/jes/article/doi/10.1210/jendso/bvaf149.1262/8299379>
28. <https://academic.oup.com/jes/article/doi/10.1210/jendso/bvaf149.1098/8299472>
29. <https://academic.oup.com/jes/article/doi/10.1210/jendso/bvaf149.1099/8299471>
30. <https://academic.oup.com/jes/article/doi/10.1210/jendso/bvaf149.1149/8299622>
31. <https://academic.oup.com/jes/article/doi/10.1210/jendso/bvaf149.1124/8299535>
32. <https://academic.oup.com/jes/article/doi/10.1210/jendso/bvaf149.1123/8299536>
33. <https://academic.oup.com/jes/article/doi/10.1210/jendso/bvaf149.1096/8297594>
34. <https://www.frontiersin.org/articles/10.3389/fresc.2024.1374850/pdf?isPublishedV2=False>
35. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10933123/>
36. <https://pmc.ncbi.nlm.nih.gov/articles/PMC12097986/>
37. <https://www.foi.gov.ph/agencies/fnri/monthly-prediabetes-data/>
38. <https://newsinfo.inquirer.net/2138555/diabetes-rising-in-ph-how-culture-and-diet-fuel-the-crisis-2>
39. <https://www.facebook.com/DOST.FNRI/posts/watch-dost-fnris-recent-data-on-pre-diabetes-among-filipinos-according-to-the-201051180617200411/>
40. <https://ijarcce.com/wp-content/uploads/2025/09/IJARCCE.2025.14820.pdf>
41. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10041290/>
42. <https://www.sciencedirect.com/science/article/pii/S2211335523002498>
43. <https://pmc.ncbi.nlm.nih.gov/articles/PMC12563305/>
44. [https://www.ijaemr.com/uploads/pdf/archivepdf/2025/IJAEMR\\_706.pdf](https://www.ijaemr.com/uploads/pdf/archivepdf/2025/IJAEMR_706.pdf)
45. <https://arxiv.org/pdf/2305.00070.pdf>
46. <https://arxiv.org/pdf/2502.15131.pdf>
47. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9742671/>
48. <https://arxiv.org/ftp/arxiv/papers/2110/2110.00918.pdf>
49. [https://dl.acm.org/doi/10.1007/978-3-031-16749-2\\_9](https://dl.acm.org/doi/10.1007/978-3-031-16749-2_9)
50. <https://pubsonline.informs.org/doi/10.1287/ijds.2024.0038>
51. <https://arxiv.org/html/2509.23665v1>
52. <https://www.semanticscholar.org/paper/On-the-appropriateness-of-Platt-scaling-in-Böken/d539ceb565d1920381072aff96e3b1f57485c298>

53. <https://www.semanticscholar.org/paper/e4c8bb2503e5d954731dcde19e49939046c72c22>
54. <https://pmc.ncbi.nlm.nih.gov/articles/PMC3277613/>
55. <http://arxiv.org/pdf/2406.11456.pdf>
56. <https://arxiv.org/pdf/2311.05418.pdf>
57. <http://www.ajnr.org/content/ajnr/32/2/219.full.pdf>
58. <https://www.blog.trainindata.com/complete-guide-to-platt-scaling/>
59. <https://gjeta.com/sites/default/files/GJETA-2025-0267.pdf>
60. <https://www.jmir.org/2025/1/e73190>
61. <https://www.sciencedirect.com/science/article/abs/pii/S1361841523000841>
62. <https://bioinform.jmir.org/2025/1/e70621/PDF>
63. <https://arxiv.org/html/2509.20565v1>
64. <https://www.sciencedirect.com/science/article/pii/S2666307421000279>
65. <https://journals.plos.org/plosone/article?id=10.1371%2Fjournal.pone.0327120>
66. <https://www.nature.com/articles/s41598-025-10136-5>
67. [https://www.arfjournals.com/image/catalog/Journals Papers/AJSS/2025/No 1 \(2025\)/AJSS 6\\_87-96.pdf](https://www.arfjournals.com/image/catalog/Journals%20Papers/AJSS/2025/No%201%20(2025)/AJSS%206_87-96.pdf)
68. <https://dl.acm.org/doi/10.1145/3745812.3745842>